



EE115 Analog Circuits

模拟电路

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Outline



- MOSFET
- SEDTRA/SMITH book page 247-276;

MOSFET

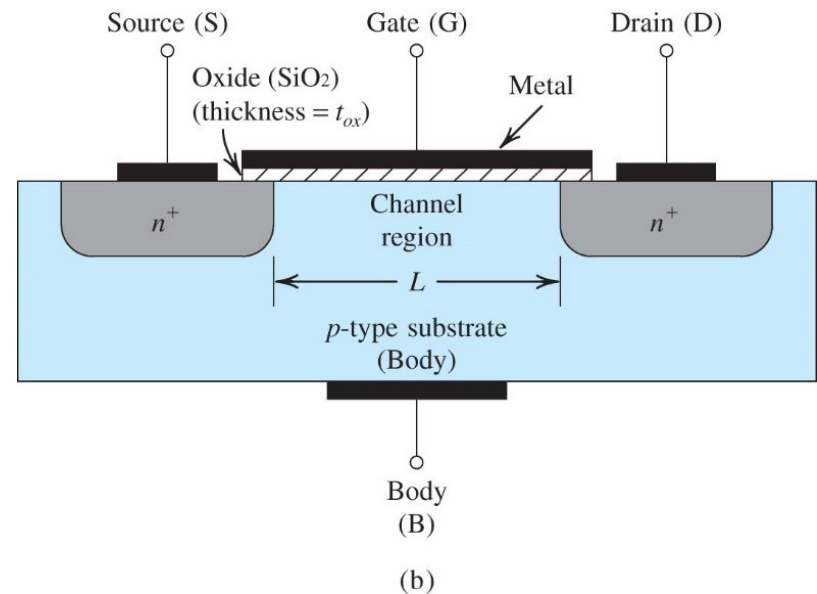
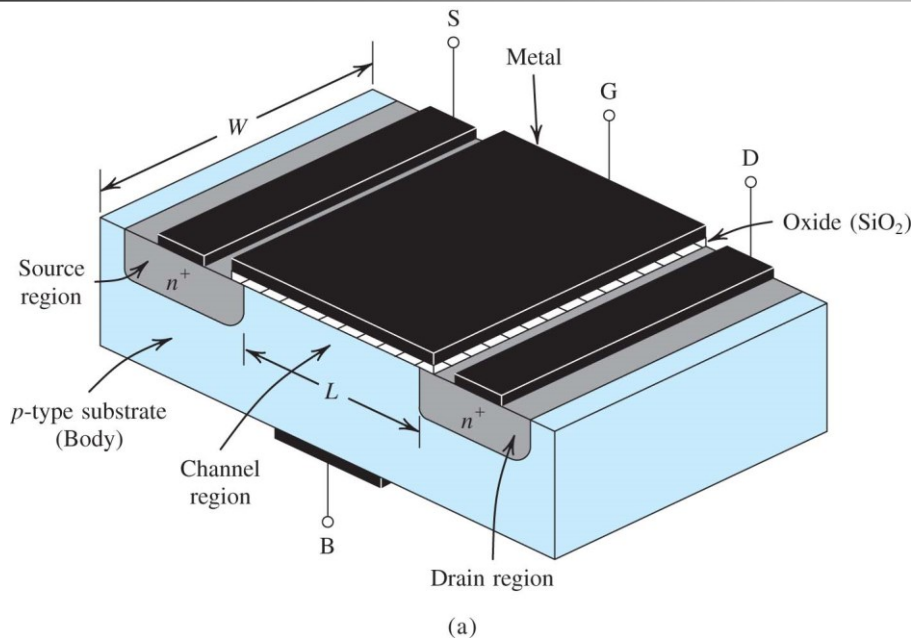


Figure 5.1 Physical structure of the enhancement-type NMOS transistor: **(a)** perspective view; **(b)** cross section. Typically $L = 0.03 \mu\text{m}$ to $1 \mu\text{m}$, $W = 0.05 \mu\text{m}$ to $100 \mu\text{m}$, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm.

- MOSFET: metal-oxide-semiconductor field effect transistor
- Typically
 - Channel length: $L \sim 3 \text{ nm}$ to $0.35 \mu\text{m}$
 - Channel width: $W \sim 0.05 \mu\text{m}$ to $100 \mu\text{m}$
 - Oxide thickness: $t_{ox} \sim 1$ to 10 nm

NMOSFET (NMOS)

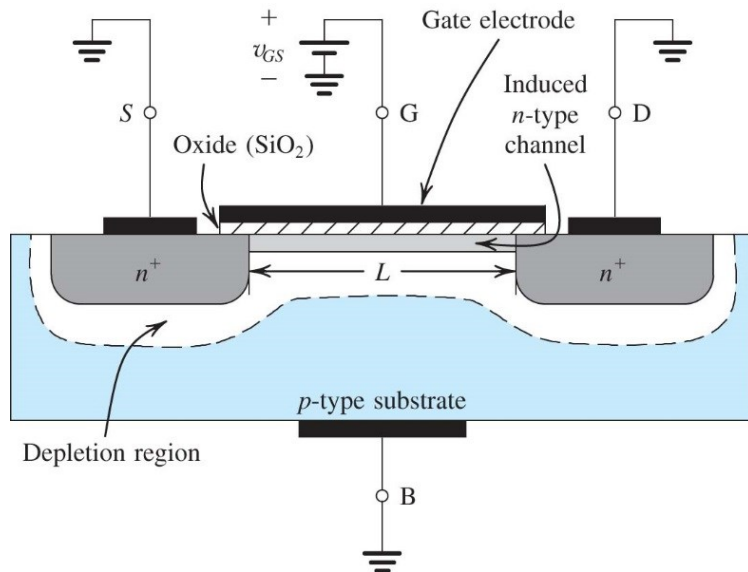


Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

- **N-channel (n-沟道) MOSFET**
 - Current conducted by e⁻s
- 3 terminal device
 - **Source (S, 源)**: n⁺ (heavily doped n-type)
 - **Drain (D, 漏)**: n⁺
 - **Gate (G, 栅)**; metal deposited on insulator above channel
- **Substrate (Body, 衬底, 体)** is a 4th terminal
 - Substrate is p-doped
- Electrons are induced in channel when a positive gate voltage is applied
- Electrons moves from Source to Drain
 - Current flows from D to S

Creating a “Channel (沟道)” for Current Flow

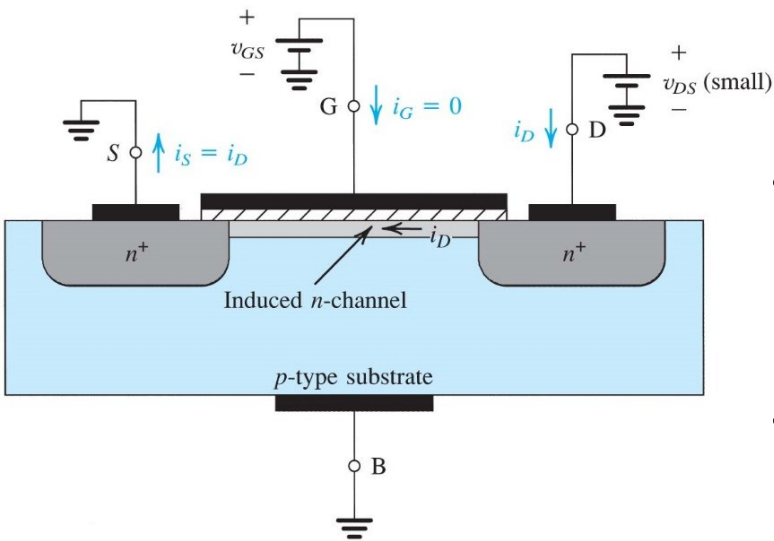


Figure 5.3 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a resistance whose value is determined by v_{GS} . Specifically, the channel conductance is proportional to $v_{GS} - V_t$, and thus i_D is proportional to $(v_{GS} - V_t)v_{DS}$. Note that the depletion region is not shown.

- MOS is a capacitor across an insulator (oxide) When a positive voltage is applied at Gate, electrons are induced under the gate.
- At “**threshold (阈值)**”, sufficient number of electrons form a "channel" between Source and Drain, forming a conductive channel.

- Total charge in the channel:

$$|Q| = C_{ox} \cdot WL \cdot (v_{GS} - V_t)$$

where $C_{ox} = \epsilon_{ox}/t_{ox}$ is oxide capacitance per unit area

- $\epsilon_{ox} = 3.9\epsilon_0 = 3.9 \times 8.854 \times 10^{-12} \text{ F/m}$
- W : gate width
- L : gate length
- V_t : Threshold voltage
- $v_{GS} - V_t \equiv v_{OV}$ is called “**Overdrive Voltage(过驱动电压)**”

Current at Small v_{DS}

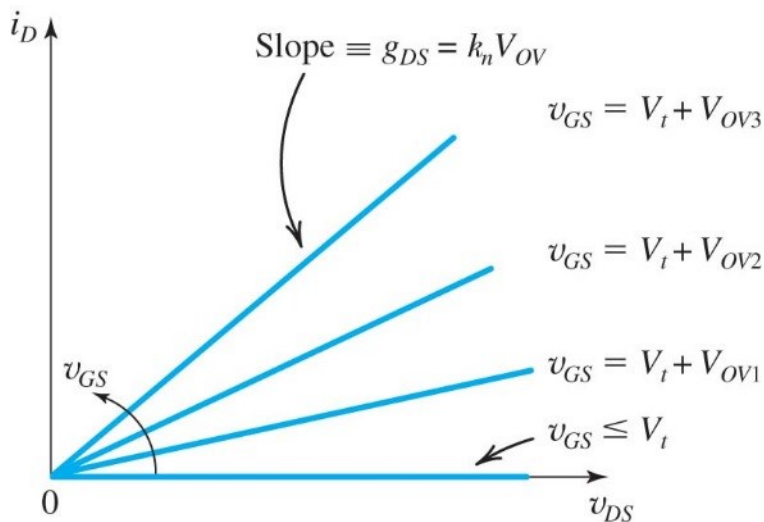


Figure 5.4 The $i_D - v_{DS}$ characteristics of the MOSFET in Fig. 5.3 when the voltage applied between drain and source, v_{DS} , is kept small. The device operates as a linear resistance whose value is controlled by v_{GS} .

- When $v_{OV} = v_{GS} - V_t > 0$, a channel is formed between Source and Drain.
- Linear charge density in channel:

$$\frac{|Q|}{\text{unit channel length}} = C_{ox} W v_{OV}$$

- Electric field along the channel:

$$|E| = \frac{v_{DS}}{L}$$

- Drain current = charge density $\times$ velocity:

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$$

- At small v_{DS} , the transistor is like a gate-controlled variable resistor

Continued

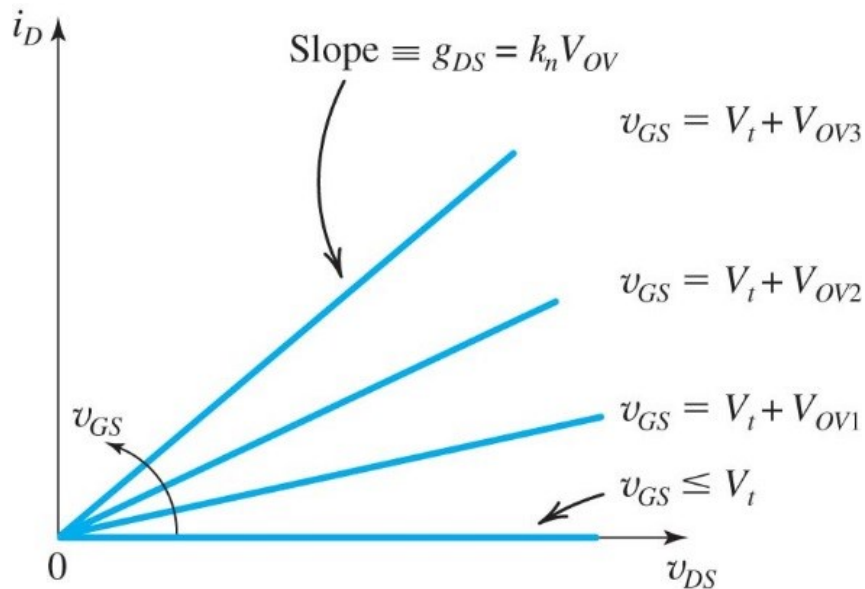


Figure 5.4 The i_D - v_{DS} characteristics of the MOSFET in Fig. 5.3 when the voltage applied between drain and source, v_{DS} , is kept small. The device operates as a linear resistance whose value is controlled by v_{GS} .

- *Transconductance parameters:*

$$k'_n = \mu_n C_{ox}$$

$$k_n = k'_n (W/L) = (\mu_n C_{ox})(W/L)$$

- *Gate-controlled resistance*

$$r_{DS} = \frac{1}{(\mu_n C_{ox})(W/L)(v_{GS} - V_t)}$$

Triode Region ($v_{DS} < v_{OV}$)

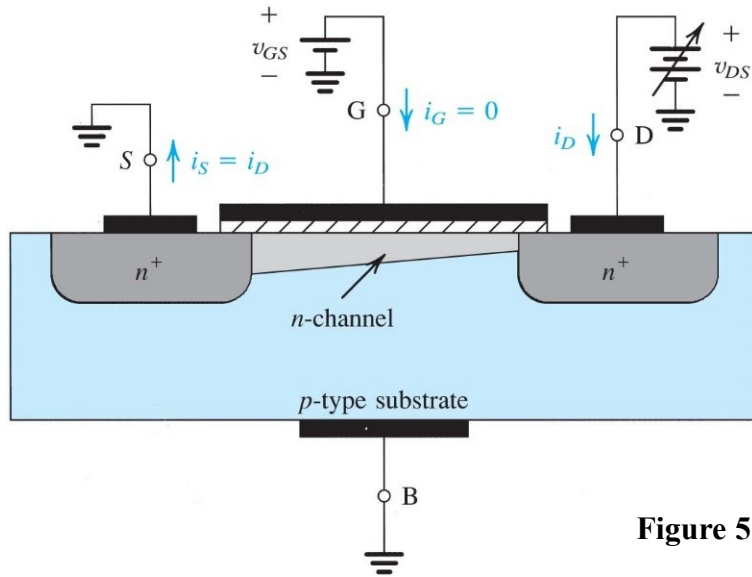
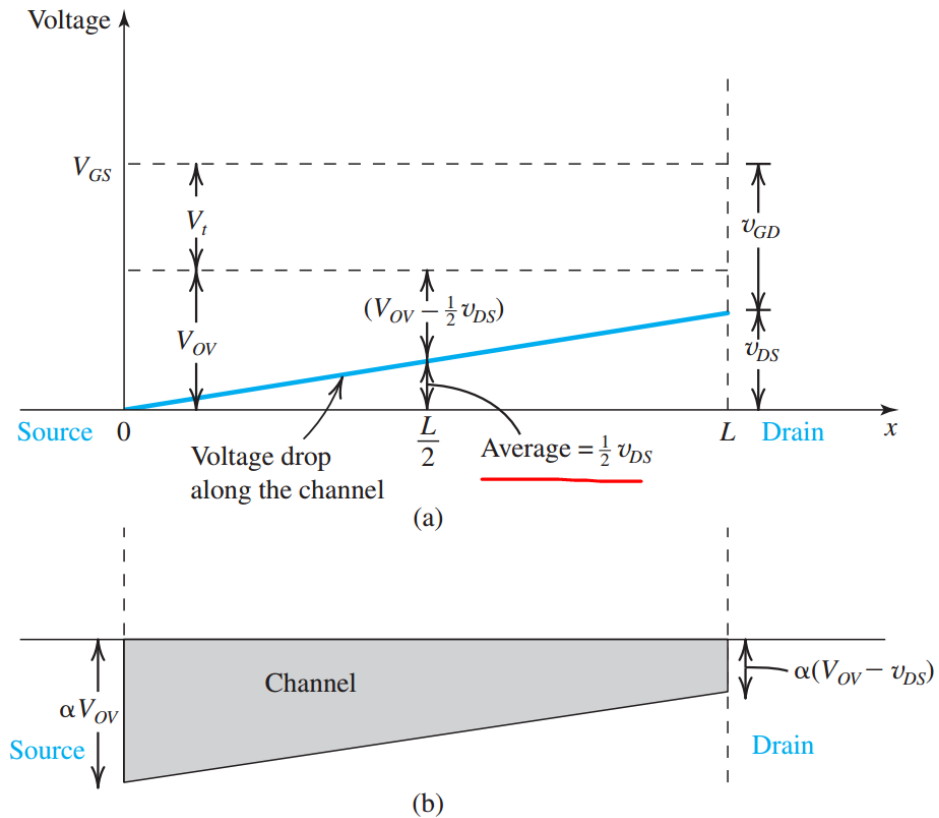


Figure 5.5

- As v_{DS} increases, the potential in the channel is no longer a constant. Assume the channel is $v(x)$:

$$i_D = k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) v_{DS} = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$



I-V curve

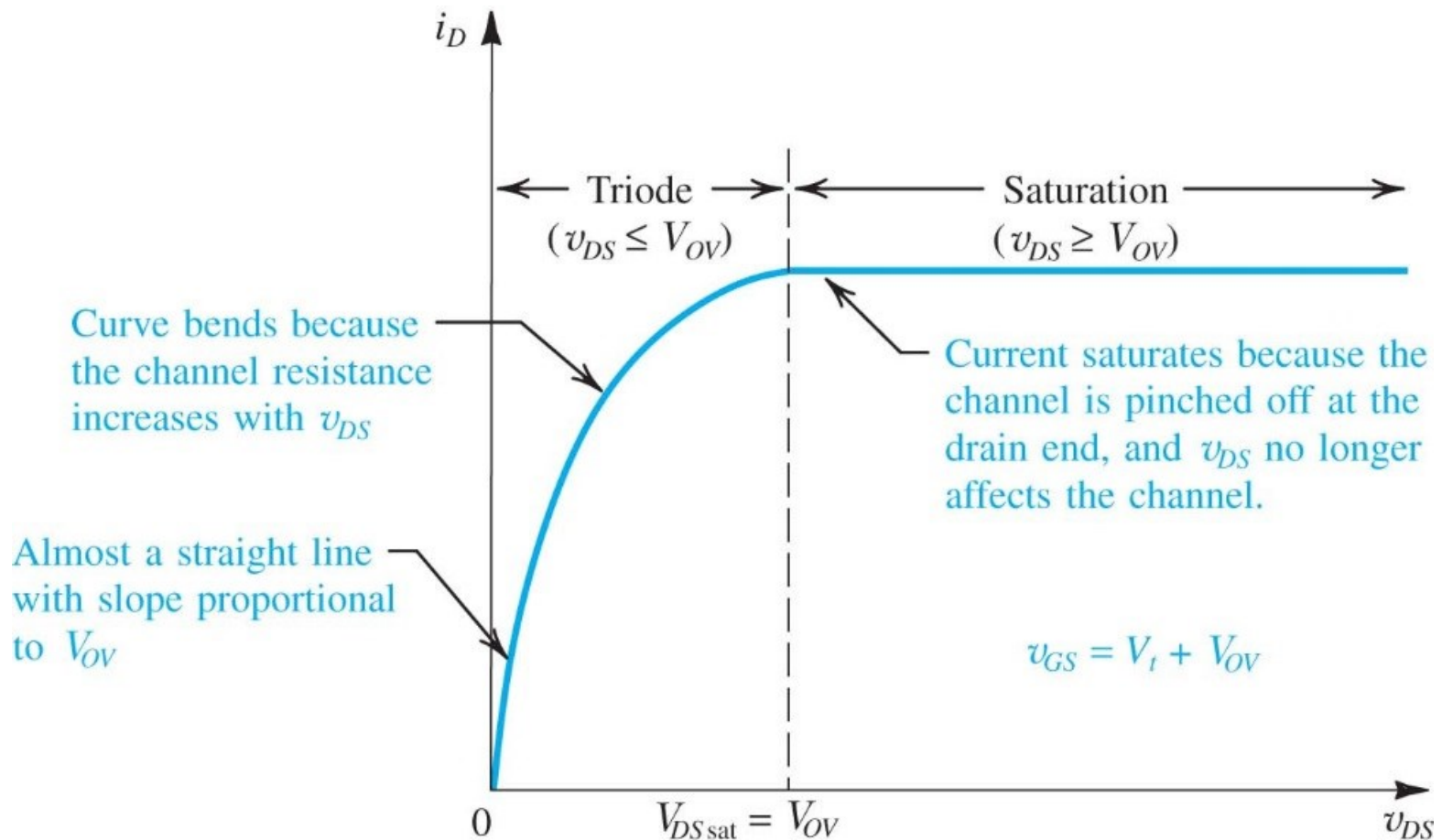


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + V_{OV}$

Pinch-Off (夹断)

The channel potential at the drain side is v_{DS} .

When $v_{DS} = v_{OV}$, the local charge density at the drain becomes zero.

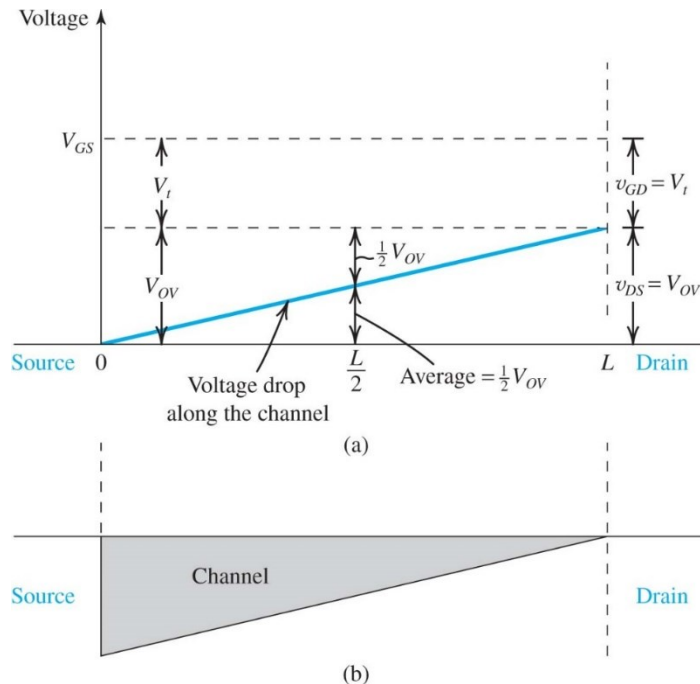


Figure 5.8

So the channel is "pinched off" near the Drain.

Once the channel is pinched off, the drain current remains constant:

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$

This region, $v_{DS} > v_{OV}$, is called "Saturation (饱和)"

Saturation Region ($v_{DS} \geq v_{OV}$)

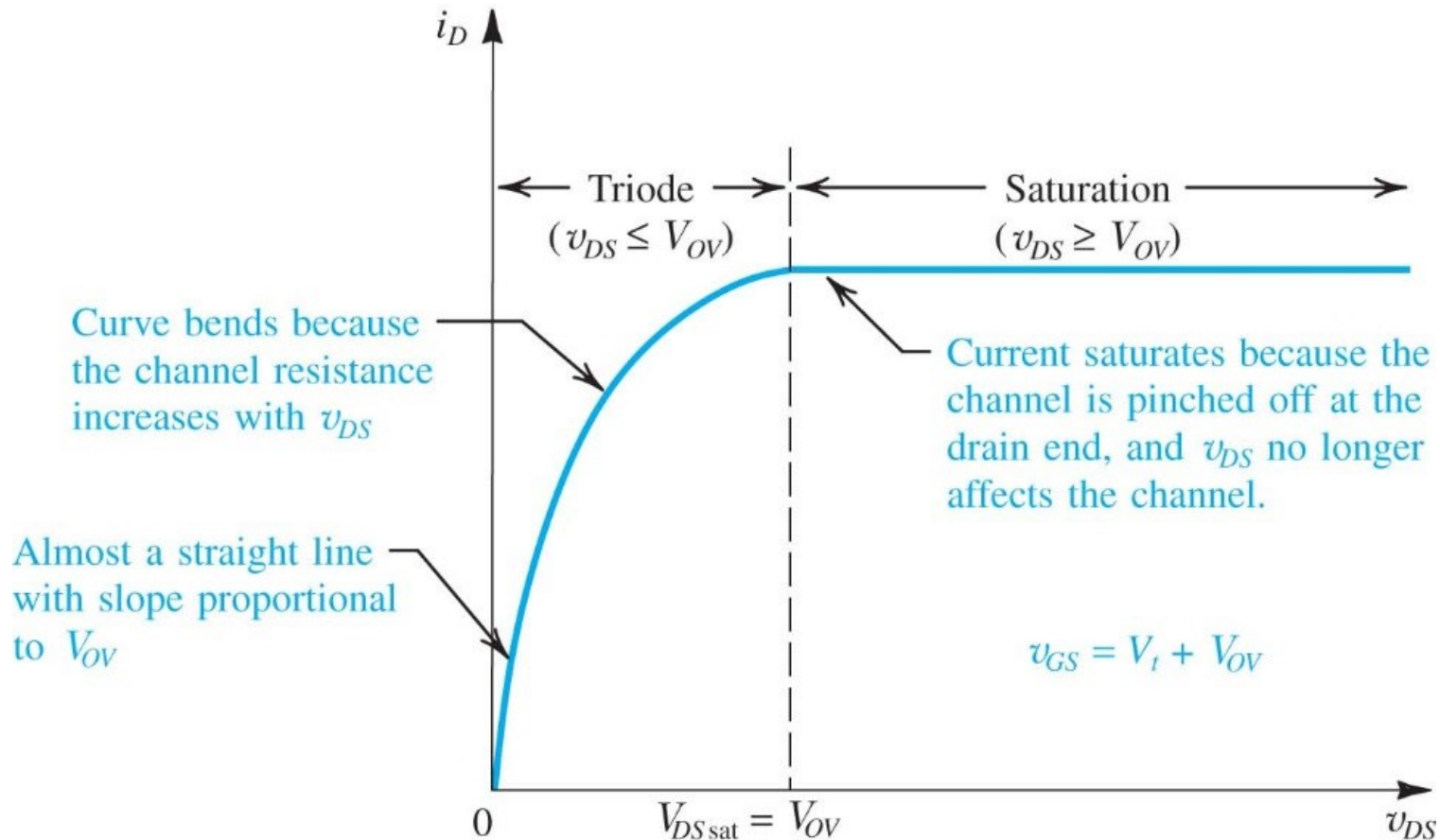


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + V_{OV}$

I-V Curves of NMOS

$$i_D = k'_n \left(\frac{W}{L} \right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right)$$

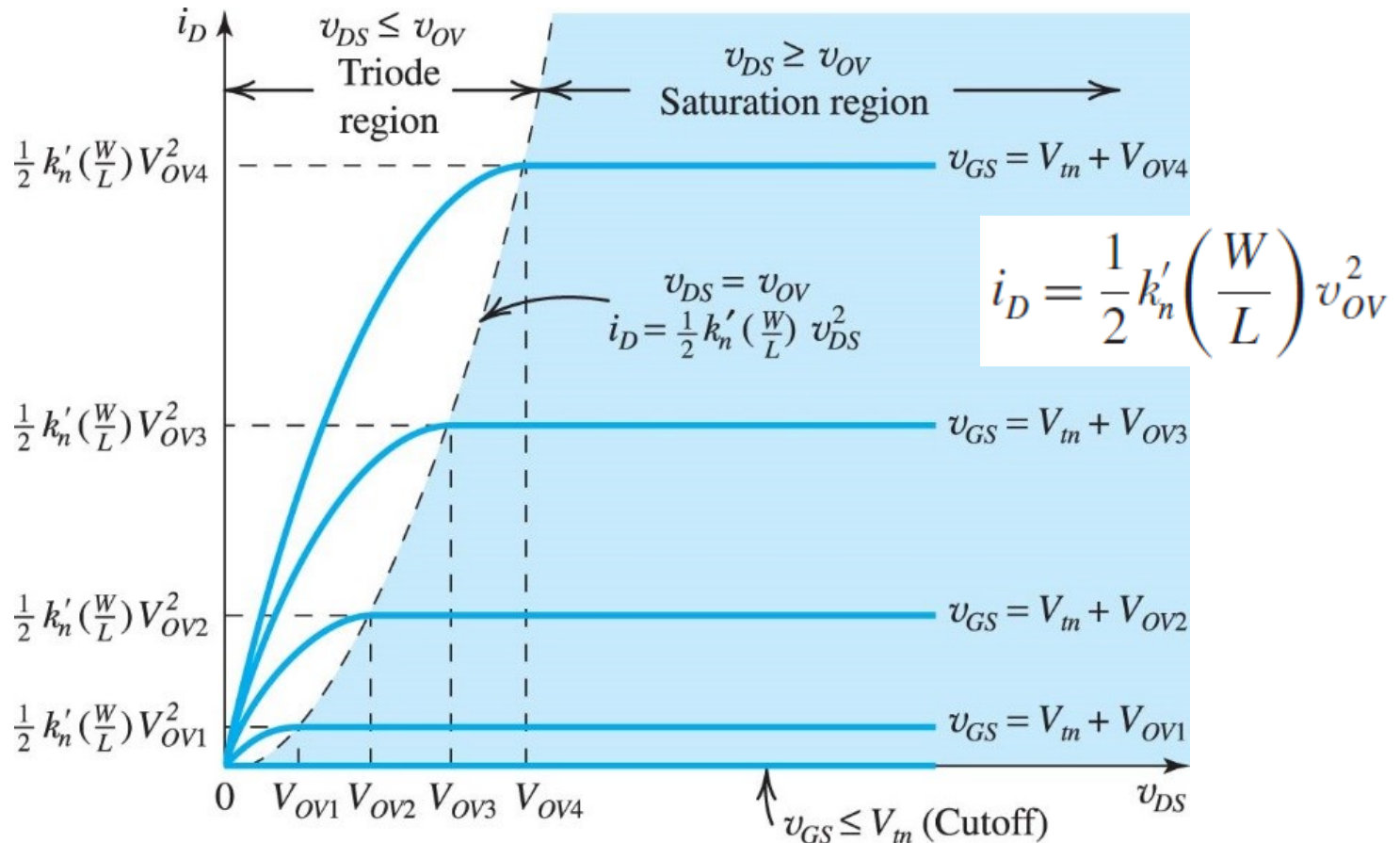
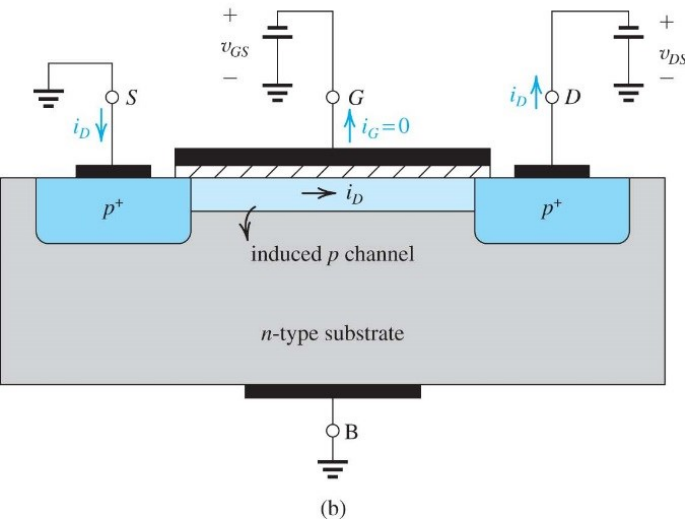
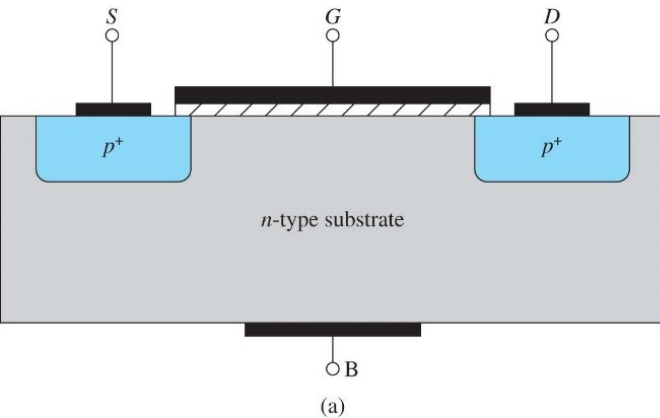


Figure 5.13 The $i_D - v_{DS}$ characteristic for an enhancement-type NMOS transistor.

PMOSFET (PMOS)



- P-channel MOSFET
 - Current conducted by holes
- 3 terminal device
 - Source (S): p+ (heavily p-type)
 - Drain (D): p+
 - Gate (G): metal deposited on insulator above channel
- Substrate (called “Body”) is a 4th terminal
 - Substrate is n-doped
- Holes is induced in channel when a negative gate voltage is applied
- Holes moves from Source to Drain
 - Current flows from S to D

CMOS (Complementary MOS)

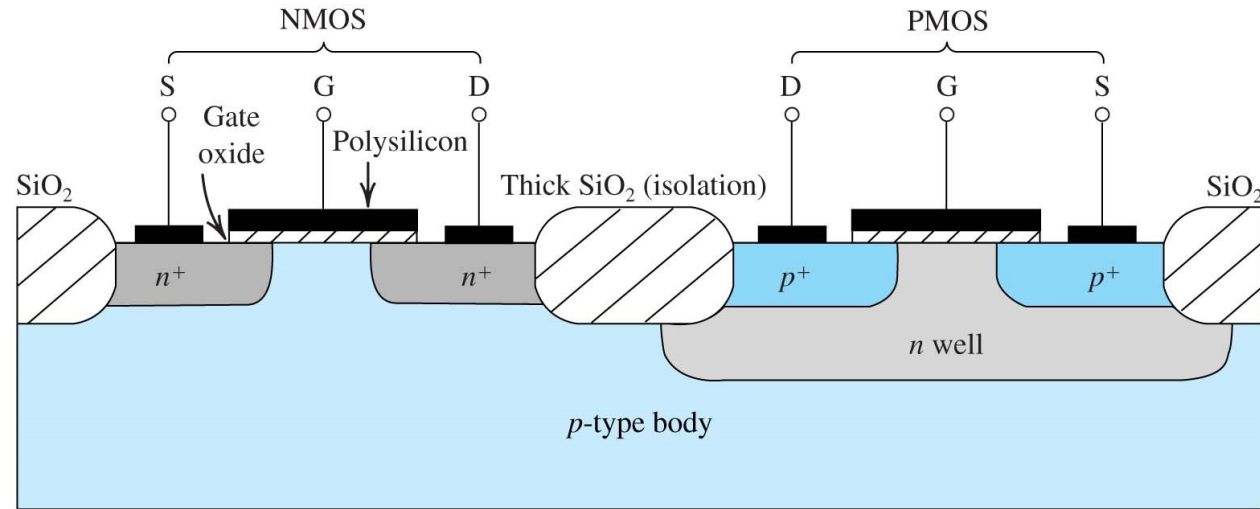
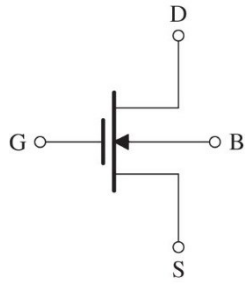


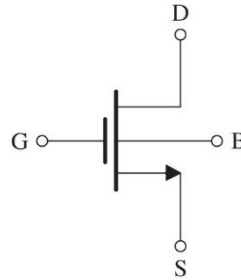
Figure 5.10 Cross section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well. Another arrangement is also possible in which an *n*-type substrate (body) is used and the *n* device is formed in a *p* well. Not shown are the connections made to the *p*-type body and to the *n* well; the latter functions as the body terminal for the *p*-channel device.

- CMOS is the prevalent IC technology today
- Since NMOS and PMOS are formed on oppositely doped substrates, one of the transistor needs to be placed in a “**well(阱)**”
- PMOS is placed in an “**n well(n阱)**” here.
- Alternatively, NMOS can be placed in p well

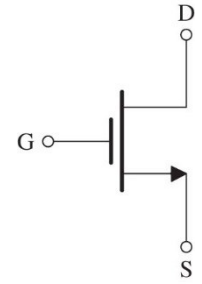
Circuit Symbol for NMOS



(a)



(b)

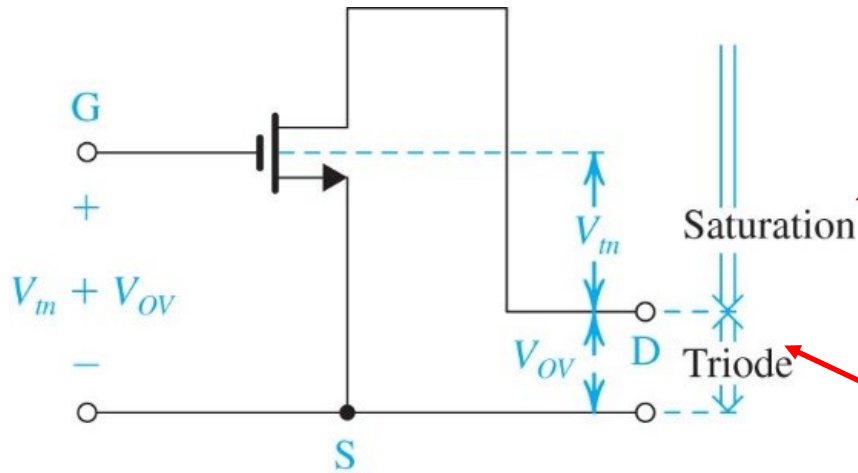


(c)

Figure 5.11 (a) Circuit symbol for the n -channel enhancement-type MOSFET. (b) Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., n channel). (c) Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

- 4 terminal including Body (Arrow pointing to channel indicating substrate is **p-type**)
- Modified circuit symbol with arrow on source (Arrow indicating direction of current flow)
- Simplified circuit symbol with body connected to source (or when the effect of the body on device operation is unimportant)
- Note in NMOS
 - » **Drain voltage is always more positive than source voltage**
 - » **Current always flows from Drain to Source**

Relative Voltage Levels of NMOS



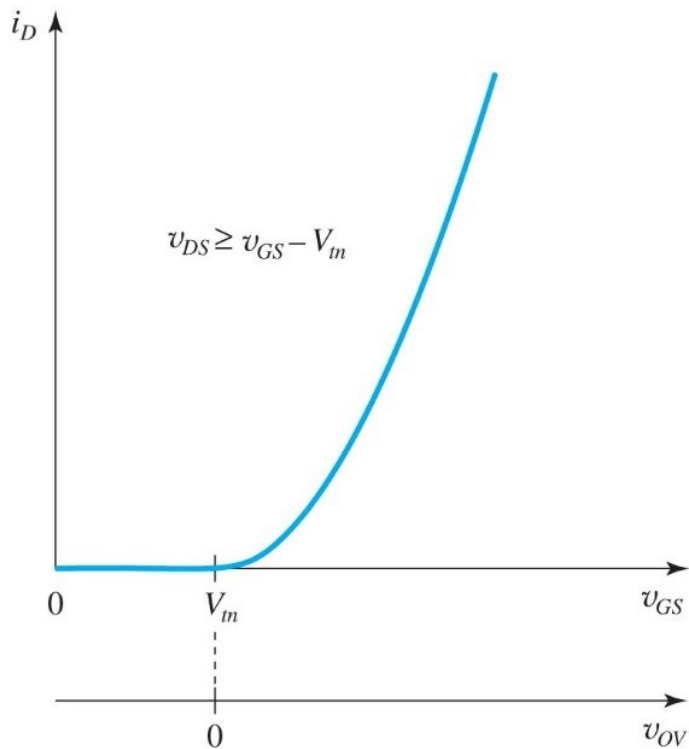
Analog Circuit (e.g., linear amplifiers) usually biased in Saturation region

Digital circuit (e.g. inverter) and power electronics (switch mode power supply) use Triode region as one of the states

V_{tn} : threshold voltage of NMOS

V_{tn} is usually fixed once a **process (technology, 工艺)** is selected

Drain Current vs Gate Voltage



- In Saturation Region

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_m)^2$$

- To experimentally determine V_{tn} :

Figure 5.14 The i_D - v_{GS} characteristic of an NMOS transistor operating in the saturation region. The i_D - v_{OV} characteristic can be obtained by simply relabeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_m$.

Finite Output Resistance due to Channel length Modulation (沟长调制)



- When $v_{DS} = v_{OV}$
Pinched-off happens

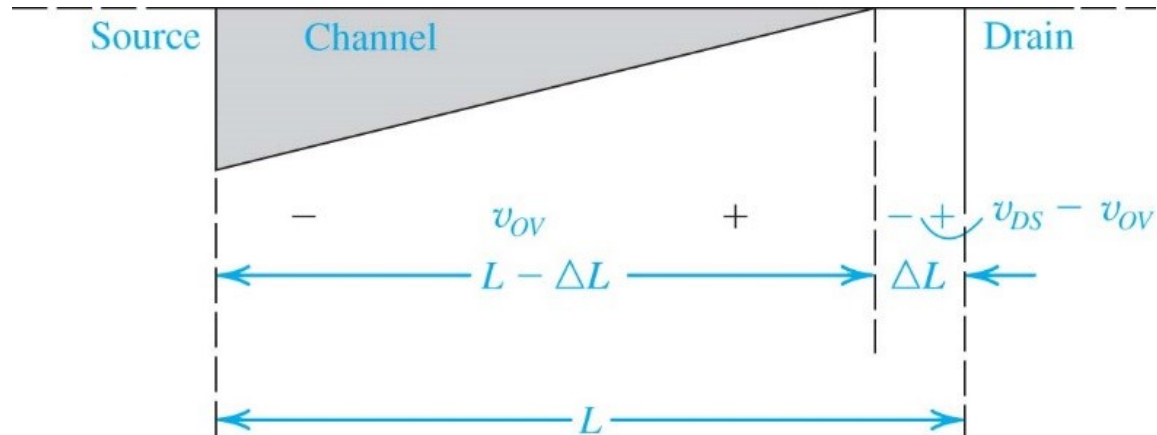


Figure 5.16 Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length (by ΔL)

- When $v_{DS} > v_{OV}$
Pinch-off point shifts to the source

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_m)^2 (1 + \lambda v_{DS})$$

Output Resistance of MOSFET

■ Output resistance of NMOS:

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} \text{ constant}}^{-1} = \left[\lambda \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_m)^2 \right]^{-1} = \frac{1}{\lambda I_D}$$

Due to CLM, the output resistance of MOSFET is not infinite, but still a large value at least for long-channel device.

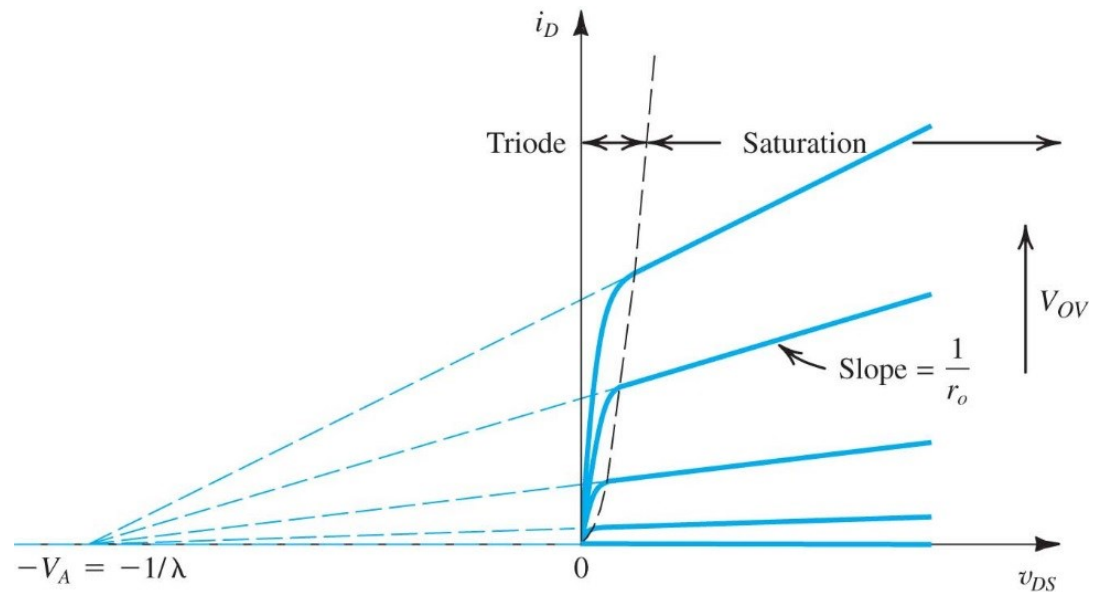


Figure 5.17 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L .

Circuit Symbol of PMOS

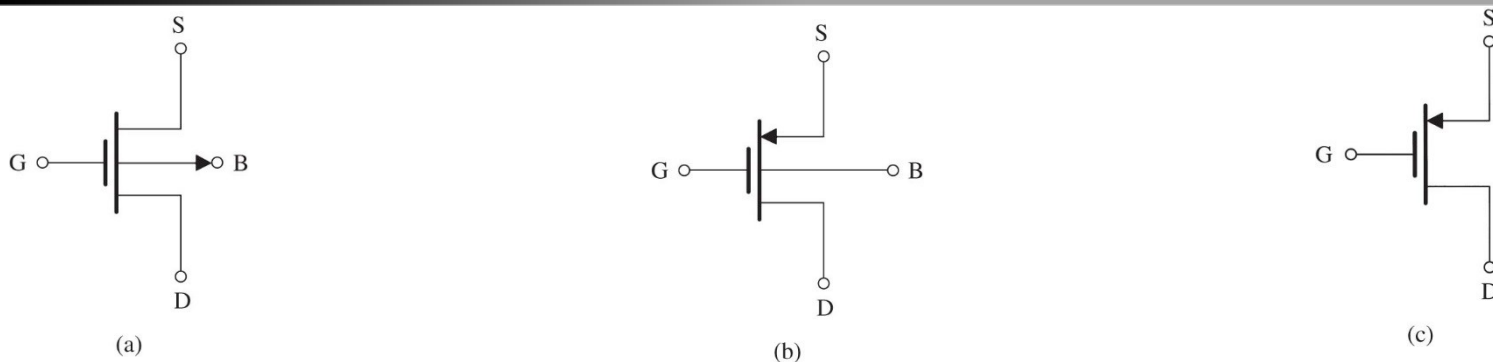
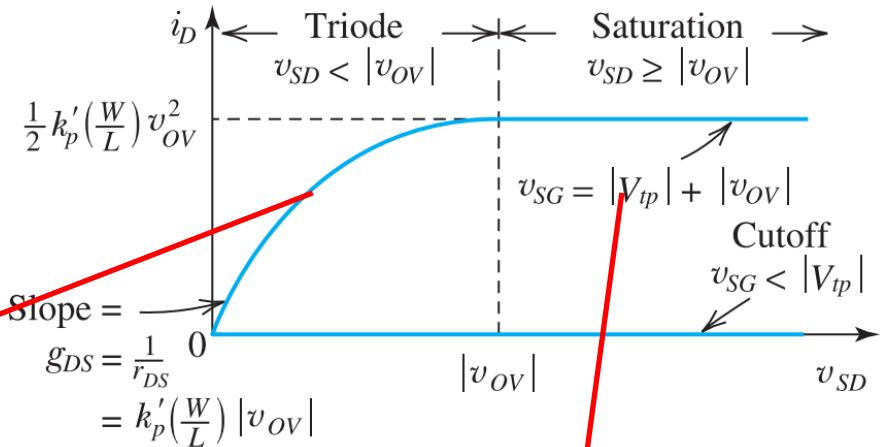
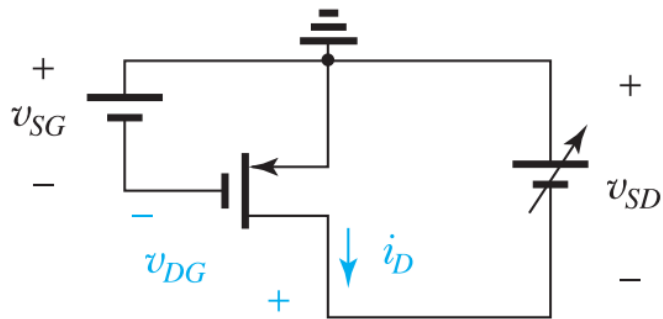


Figure 5.19 (a) Circuit symbol for the p -channel enhancement-type MOSFET. (b) Modified symbol with an arrowhead on the source lead. (c) Simplified circuit symbol for the case where the source is connected to the body.

- 4 terminal including Body (Arrow pointing **from** channel indicating substrate is **n-type**)
- Modified circuit symbol with arrow on source (Arrow indicating direction of current flow)
- Simplified circuit symbol with body connected to source (or when the effect of the body on device operation is unimportant)
- Note in PMOS
 - » Source voltage is always more positive than Drain voltage
 - » Current always flows from Source to Drain
 - » **Source is usually drawn on top so current flows downward (convention)**

PMOS I-V Equations



• Triode Region

■ Saturation Region

Relative Voltage Levels of PMOS

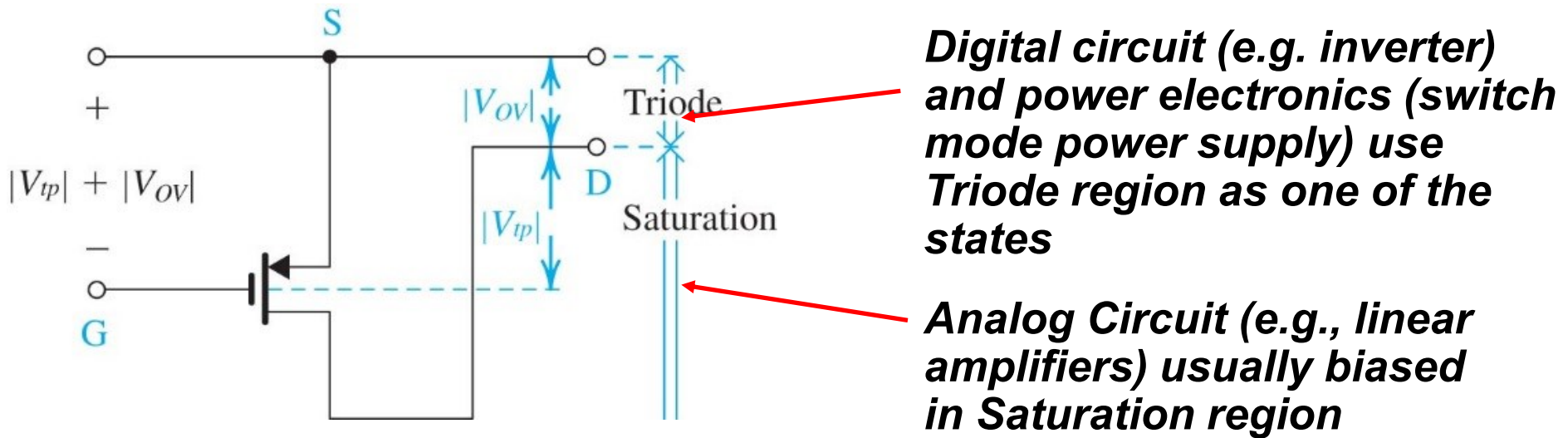


Figure 5.20 The relative levels of the terminal voltages of the enhancement-type PMOS transistor for operation in the triode region and in the saturation region.

V_{tp} : threshold voltage of PMOS

V_{tp} is usually fixed once a **process (technology, 工艺)** is selected

Diode-Connected Transistor

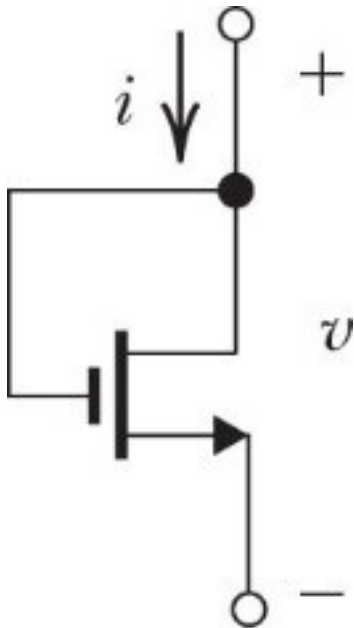


Figure 5.22

- With Gate connected to Drain, it becomes a 2-terminal device.
- This is called diode-connected transistor.
- In this configuration, the transistor is always in Saturation.

Example Circuit (1)

- Design problem:

Determine R_S and R_D such that the NMOS is biased at $I_D = 0.4\text{mA}$ and $V_D = 0.5\text{V}$. The NMOS has $V_t = 0.7\text{V}$, $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $L = 1 \mu\text{m}$ and $W = 32 \mu\text{m}$.

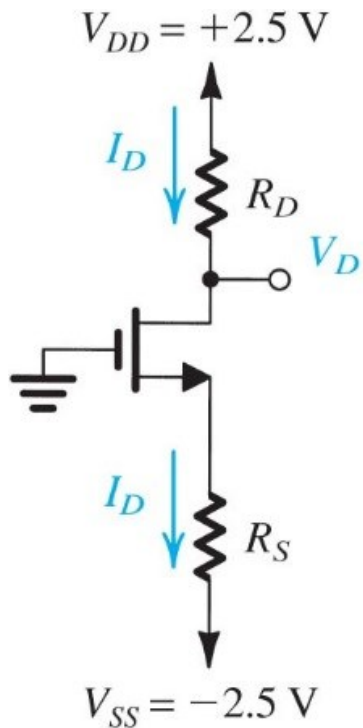


Figure 5.21 Circuit for Example 5.3.

Example Circuit (2)

- DC operating point:

Analyze the circuit to determine the voltages at all nodes and the currents through all branches. Let $V_{tn} = 1 \text{ V}$ and $k_n' (W/L) = 1 \text{ mA/V}^2$. Neglect the channel length modulation effect.

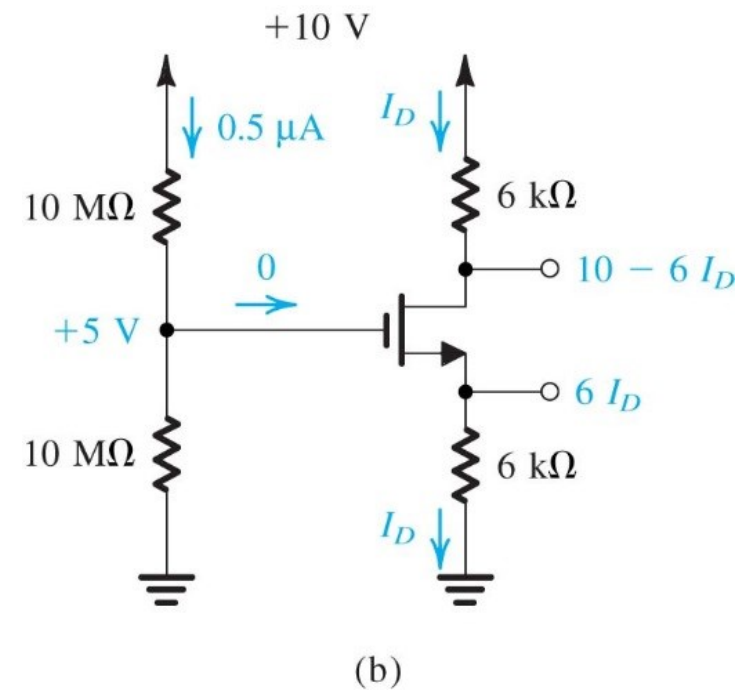
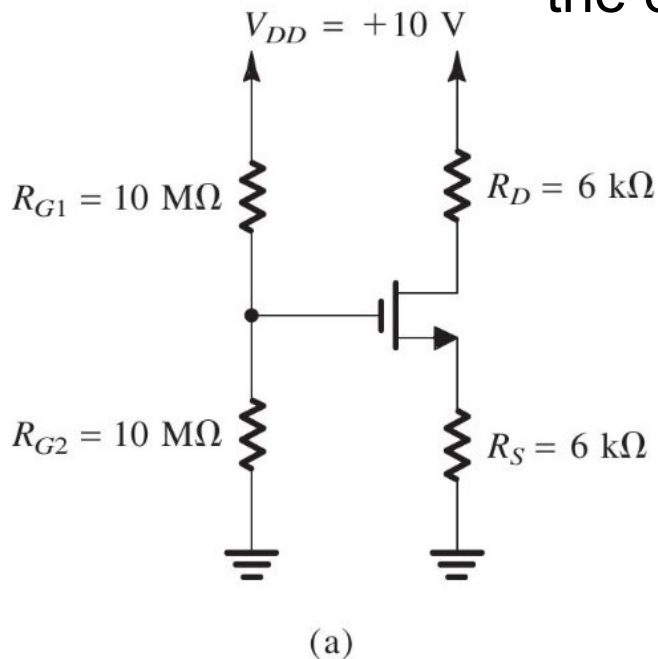


Figure 5.24 (a) Circuit for Example 5.6. (b) The circuit with some of the analysis details shown.

solution

Solution

Since the gate current is zero, the voltage at the gate is simply determined by the voltage divider formed by the two 10-M Ω resistors,

$$V_G = V_{DD} \frac{R_{G2}}{R_{G2} + R_{G1}} = 10 \times \frac{10}{10 + 10} = +5 \text{ V}$$

With this positive voltage at the gate, the NMOS transistor will be turned on. We do not know, however, whether the transistor will be operating in the saturation region or in the triode region. We shall assume saturation-region operation, solve the problem, and then check the validity of our assumption. Obviously, if our assumption turns out not to be valid, we will have to solve the problem again for triode-region operation.

Refer to Fig. 5.24(b). Since the voltage at the gate is 5 V and the voltage at the source is $I_D (\text{mA}) \times 6 (\text{k}\Omega) = 6I_D (\text{V})$, we have

$$V_{GS} = 5 - 6I_D$$

Thus I_D is given by

$$\begin{aligned} I_D &= \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_m)^2 \\ &= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2 \end{aligned}$$

which results in the following quadratic equation in I_D :

$$18I_D^2 - 25I_D + 8 = 0$$

This equation yields two values for I_D : 0.89 mA and 0.5 mA. The first value results in a source voltage of $6 \times 0.89 = 5.34 \text{ V}$, which is greater than the gate voltage and does not make physical sense as it would imply that the NMOS transistor is cut off. Thus,

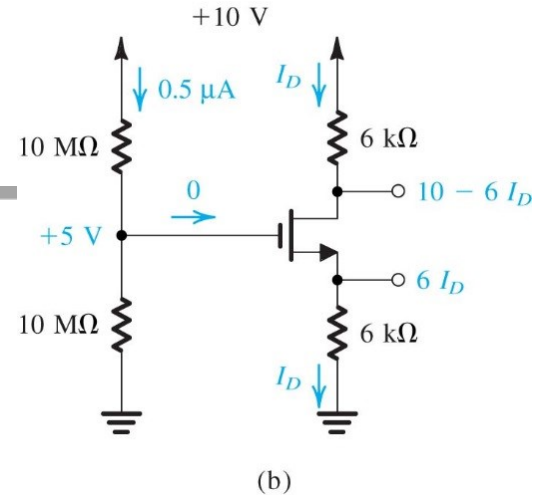
$$I_D = 0.5 \text{ mA}$$

$$V_S = 0.5 \times 6 = +3 \text{ V}$$

$$V_{GS} = 5 - 3 = 2 \text{ V}$$

$$V_D = 10 - 6 \times 0.5 = +7 \text{ V}$$

Since $V_D > V_G - V_m$, the transistor is operating in saturation, as initially assumed.



Homework3-due April. 8th 11p.m.



SEDRA/SMITH Book chapter 3 problem 3.13,
3.20, 3.24, chapter 5 problem 5.2, 5.7, 5.18, 5.20